

DNV-1.0: Contracted Observables for Admissible Representation and Stable Interpretation in Materials Informatics

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Abstract

Interpretability in materials machine learning is often compromised when preprocessing alters what an “observable” means, while headline metrics barely move. Scaling, casting, and imputation can shift units, bounds, or reference states, such that a feature no longer matches the physical quantity it purports to encode. Data Nexus Vista (DNV-1.0) treats each observable as a *contracted object*: a value paired with an explicit semantic contract. The contract specifies units, dimensionality, admissible bounds, and required context, and constrains how the observable may be transformed. Representations are built only from contract-admissible operations, making semantic drift a detectable violation rather than

a silent failure. All transformations are recorded in a provenance graph, allowing any value to be reconstructed and audited. Instead of pointwise outlier deletion, DNV-1.0 applies reversible, distribution-constrained conditioning and generates an *admissible family* of dataset variants —a controlled uncertainty set over realistic preprocessing choices that stays within declared physical constraints. Interpretations are treated as conditional statements and are reported only when they remain stable across this admissible family. On 258 DFT-computed spinel cathodes, DNV-1.0 predicts ion migration barriers with a mean absolute error of ~ 0.1 eV and preserves leading-attribution stability under admissible changes of representation. Mechanistic claims then become falsifiable stability statements: they are credible only insofar as they survive declared, auditable changes of representation.

Keywords: Materials informatics; Representation stability; Data provenance; Physically admissible modeling; Interpretable machine learning

1 Distribution-Constrained Data Conditioning

(see SI for implementation details). Not implemented yet.

Outlier handling and dataset conditioning are usually pointwise: samples are removed, clipped, or downweighted using thresholds, distances, or anomaly scores. Such edits can be locally defensible but globally opaque. Moving or deleting a few points can shift moments, correlations, and feature rankings in ways that are difficult to predict. In heterogeneous materials datasets, apparent “outliers” often reflect real physics—metastability, uncommon coordination, or extreme electronic structure. Pointwise filtering can therefore erase plausible regimes and destabilize downstream interpretation.

DNV-1.0 treats conditioning as an explicit epistemic intervention constrained by observable semantics. We implement this as the *Semantic Envelope for Admissible Conditioning* (SEAL) module. SEAL does not label observations as “bad.” It searches for *minimal*, contract-feasible transformations that move the empirical distribution into an admissible envelope $\mathcal{Q}(\theta)$ defined by user-declared, physically grounded tolerances. We treat $\mathcal{Q}(\theta)$ as an uncertainty set: the family of empirical distributions consistent with observable contracts and declared physical expectations. Conditioning may reduce pathological leverage, but it may not silently change what an observable means. SEAL operates on X only. Rules that depend on targets, residuals, or label-derived errors are prohibited unless explicitly declared as part of a measurement model. To prevent information leakage, SEAL is fit on training data only (or within each cross-validation fold) and then applied to held-out splits.

Admissibility constraints.

Admissibility is represented by a finite set of constraint families evaluated on empirical statistics and enforced jointly with observable contracts. Let P_X denote the empirical distribution induced by dataset X . Each constraint family j defines a nonnegative residual $r_j(P_X)$ and a tolerance θ_j , declared by the user (or set to conservative defaults). The admissible envelope is

$$\mathcal{Q}(\theta) := \{P : r_j(P) \leq \theta_j \ \forall j, \text{ and all observable contracts are satisfied}\}. \quad (1)$$

- **Quantile and tail-mass constraints:** tolerances on robust quantiles (e.g., $q_{0.05}$, $q_{0.50}$, $q_{0.95}$) and on tail mass beyond declared thresholds. These constraints regulate leverage while preserving physically plausible tails. Optional trimmed-moment bounds (skewness/kurtosis) are enabled only above a minimum sample size, and their uncertainty is reported.
- **Physical bounds:** ranges derived from prior knowledge, measurement limits, or computational protocols, reconciled with contracts (units, reference state, and admissible domain).
- **Latent reference constraints (optional):** proximity scores in a low-dimensional embedding learned from a declared reference corpus (versioned and hash-logged). By default these scores are diagnostic. If used in conditioning, they act only as capped soft preferences and never override contracts or “correct” observations. Distant points are treated as potentially informative, high-influence cases. If the reference corpus overlaps the study dataset, embeddings are learned by cross-fitting to prevent leakage.

Contract-feasible transformation search.

SEAL selects transformations from a low-dimensional, physically interpretable family $\mathcal{T}$ and rejects any candidate that violates semantic feasibility. We solve:

$$\min_{T \in \mathcal{T}} \Omega(T; X) \quad \text{s.t.} \quad \mathcal{C}(T; X) = 0, \quad d(P_{T(X)}, \mathcal{Q}(\theta)) \leq \tau, \quad (2)$$

where $\mathcal{C}(T; X) = 0$ encodes hard contract feasibility (units, bounds, reference state, missingness semantics, and provenance consistency). We instantiate the admissibility check through a weighted norm of constraint violations,

$$d(P_X, \mathcal{Q}(\theta)) := \|[r(P_X) - \theta]_+\|_{w,p}, \quad (3)$$

with $r(P_X) = (r_1(P_X), \dots, r_J(P_X))$ and weights w declared by the user or set to conservative defaults. We take $p = 1$ by default for decomposability and per-constraint attribution. The distortion functional $\Omega(T; X)$ enforces minimality in contract-normalized units (per-feature scales derived from contract ranges or robust scales such as IQR). It also enforces per-feature and per-sample budgets to prevent feasibility from being achieved by rewriting a small subset of observations. The objective ranks feasible candidates; admissibility is determined solely by the declared constraints and contracts.

The family $\mathcal{T}$ is intentionally limited. Defaults include contract-respecting smooth clamping (soft Winsorization) and bounded Winsorization. More flexible monotone maps are optional and restricted to a small, auditable parametric family with bounded curvature. Affine shifts and scalings are permitted only when they correspond to declared unit conversions or reference transformations in the observable contract. For transparency and reproducibility, SEAL uses bounded search and coordinate-wise refinement (with logged random seeds) rather than high-dimensional unconstrained optimization. When multiple feasible candidates exist, SEAL returns a diversified subset spanning admissibility margin and distortion cost (Pareto filtering with parameter-space diversity and a fixed cap on K).

Admissible families, not single datasets.

SEAL returns an admissible family $\{T_k(X)\}_{k=1}^K$ of distinct, contract-feasible variants satisfying $\mathcal{C}(T_k; X) = 0$ and $d(P_{T_k(X)}, \mathcal{Q}(\theta)) \leq \tau$. The family is an explicit uncertainty set over realistic preprocessing choices. Downstream claims are reported only if they remain stable across this family. Unless otherwise stated, models are retrained for each $T_k(X)$ using identical splits, seeds, and training budgets, so stability reflects representation choices rather than stochastic optimization. SEAL itself is fit on training data only (or per CV fold), and the learned family is then applied to held-out splits.

Relation to robust statistics.

Classical approaches (Z-score filtering, Mahalanobis distance, Huber losses, robust PCA, extreme-value models) target robustness under implicit distributional assumptions. They typically do not enforce units, reference states, or provenance constraints, and they can suppress physically meaningful tails. SEAL differs by:

1. enforcing admissibility through explicit, user-declared constraints coupled to observable contracts;
2. regulating tail behavior while bounding semantic distortion, rather than deleting points;
3. producing a family of admissible variants and using that family to quantify inference stability;
4. linking conditioning directly to descriptor, attribution, and prediction stability.

Distributional influence.

For each sample x_i , we compute a distributional influence score,

$$\delta_i = d(P_X, \mathcal{Q}(\theta)) - d(P_{X \setminus x_i}, \mathcal{Q}(\theta)), \quad (4)$$

and refer to δ_i as *Envelope Violation Influence (EVI)*. EVI is reported both in aggregate and resolved by constraint family, attributing leverage to specific admissibility pressures (e.g., tail mass versus physical bounds). Exact leave-one-out is used for small datasets. Larger collections use a block jackknife by default, with optional first-order approximations that operate directly on the constraint residuals $r_j(P_X)$. High EVI does not imply error. It flags observations whose presence materially changes admissibility under the declared envelope. SEAL reports EVI together with the active

constraint set, so influence is attributable to specific envelope pressures rather than to a generic anomaly score.

Stability diagnostics.

Given admissible variants $\{T_k\}$, descriptor importance $\phi_f^{(T_k)}$ and predictive loss $\mathcal{L}^{(T_k)}$ are evaluated as:

$$\sigma_f^2 = \text{Var}_k(\phi_f^{(T_k)}), \quad \Delta_{\text{pred}} = \text{Var}_k(\mathcal{L}^{(T_k)}). \quad (5)$$

SEAL also reports rank-based stability (Kendall τ , top- k overlap, and sign consistency) to separate semantic sensitivity from scale artifacts. Pointwise filtering can reduce predictive loss while inflating σ_f^2 , indicating that attributions have become contingent on preprocessing. SEAL is designed to favor the opposite regime: reduced leverage pathologies with bounded semantic distortion and interpretations that remain stable across admissible variants. To avoid “stability by triviality,” we report both dispersion and magnitude (e.g., median $|\phi_f^{(T_k)}|$) and flag cases where stability arises because all effects are uniformly weak.

Epistemic transparency.

All constraints, weights, tolerances, candidate transformations, divergence terms, distortion budgets, and random seeds are stored as immutable metadata. Envelope parameters are either pre-declared or treated as explicit hyperparameters tuned within training-only procedures and reported as such. Each member of the admissible family is reconstructible, with feasibility checks and constraint residuals recorded. Conditioning is thereby elevated from a hidden heuristic to an auditable analytical layer that protects descriptor validity.